INTERNATIONAL MONETARY FUND

Fear and Shortages: Identifying Geopolitical Risk and Supply Disruptions in Energy Markets

Prepared by Patricia Gomez-Gonzalez, Régis Kouassi, Jianhong Liu, Joseph Moussa, Jean-Marc Natal, Andrea Paloschi, and Evgenia Weaver

WP/26/218

IMF Working Papers **describe research in progress by the author(s) and are published to elicit comments and to encourage debate.**
The views expressed in IMF Working Papers are those of the author(s) and do not necessarily represent the views of the IMF, its Executive Board, or IMF management.

2026
OCT

WORKING PAPER

©2026 International Monetary Fund WP/26/218

IMF Working Paper
Research Department

Fear and Shortages: Identifying Geopolitical Risk and Supply Disruptions in Energy Markets

Prepared by Patricia Gomez-Gonzalez, Régis Kouassi, Jianhong Liu, Joseph Moussa, Jean-Marc Natal, Andrea Paloschi, and Evgenia Weaver

Authorized for distribution by Petya Koeva Brooks
October 2026

***IMF Working Papers* describe research in progress by the author(s) and are published to elicit comments and to encourage debate.** The views expressed in IMF Working Papers are those of the author(s) and do not necessarily represent the views of the IMF, its Executive Board, or IMF management.

ABSTRACT: This paper proposes a new high-frequency structural vector autoregression model that can disentangle the drivers of oil and natural gas prices in real time. In particular, it separately identifies actual supply disruptions from geopolitical-risk shocks that raise expectations of future supply shortfalls. Local projection analysis shows that macroeconomic effects differ across shocks. Oil supply shocks significantly affect aggregate prices and unemployment, usual central banks' policy targets, whereas geopolitical supply risk shocks have only short-lived effects on prices and no effect on unemployment. All gas shocks generate persistent effects on the price level, but muted effects on unemployment. Historical price decompositions using the high-frequency model during the Russia-Ukraine war and the 2026 Middle East war show that energy prices rise due to geopolitical risk before supply disruptions materialize. Even after disruptions occur, fears of additional shortages continue to exert upward pressure on prices. These findings highlight the importance for policymakers to distinguish realized supply disruptions from geopolitical supply risk shocks.

JEL Classification Numbers:	C32, C36, Q41, Q43
Keywords:	oil; liquefied natural gas; geopolitical risk; monetary policy

Authors' email addresses:	pgomez-gonzalez@imf.org, kkouassi@imf.org, jliu349@jh.edu, jmm2449@columbia.edu, jnatal@imf.org, apaloschi@imf.org, eweaver@imf.org

WORKING PAPERS

Fear and Shortages: Identifying Geopolitical Risk and Supply Disruptions in Energy Markets

Prepared by Patricia Gomez-Gonzalez, Régis Kouassi, Jianhong Liu, Joseph Moussa, Jean-Marc Natal, Andrea Paloschi, and Evgenia Weaver [1]

[1] The authors would like to thank Petya Koeva Brooks, Deniz Igan, and Silvana Tenreyro for comments and suggestions. The authors declare that they did not receive external funding for this research and that there are no conflicts of interest.

Fear and Shortages: Identifying Geopolitical Risk and Supply Disruptions in Energy Markets

Patricia Gomez-Gonzalez[*] Régis Kouassi[†] Jianhong Liu[‡]
Joseph Moussa[§] Jean-Marc Natal[¶] Andrea Paloschi[||] Evgenia Weaver[**]

This version: September 29, 2026

Abstract

This paper proposes a new high-frequency structural vector autoregression model that can disentangle the drivers of oil and natural gas prices in real time. In particular, it separately identifies actual supply disruptions from geopolitical-risk shocks that raise expectations of future supply shortfalls. Local projection analysis shows that macroeconomic effects differ across shocks. Oil supply shocks significantly affect aggregate prices and unemployment, usual central banks' policy targets, whereas geopolitical supply risk shocks have only short-lived effects on prices and no effect on unemployment. All gas shocks generate persistent effects on the price level, but muted effects on unemployment. Historical price decompositions using the high-frequency model during the Russia-Ukraine war and the 2026 Middle East war show that energy prices rise due to geopolitical risk before supply disruptions materialize. Even after disruptions occur, fears of additional shortages continue to exert upward pressure on prices. These findings highlight the importance for policymakers to distinguish realized supply disruptions from geopolitical supply risk shocks.

JEL classification: C32, C36, Q41, Q43
Keywords: oil, liquefied natural gas, geopolitical risk, monetary policy

[*]*Research Department, International Monetary Fund. Email: pgomez-gonzalez@imf.org*
[†]*Research Department, International Monetary Fund. Email: kkouassi@imf.org*
[‡]*Economics Department, Johns Hopkins University. Email:jliu349@jh.edu*
[§]*Participated while at the IMF's Research Department. Current email: jmm2449@columbia.edu.*
[¶]*Research Department, International Monetary Fund. Email: jnatal@imf.org*
[||]*Research Department, International Monetary Fund. Email: apaloschi@imf.org*
[**]*Research Department, International Monetary Fund. Email: eweaver@imf.org*

1 Introduction

In March 2026, the war in the Middle East effectively closed the Strait of Hormuz, locking in the Gulf about 20 percent of world oil and liquefied natural gas (LNG) consumption. The International Energy Agency (IEA) described it as the largest supply disruption since the 1970s. After peaking at $126 per barrel in March, oil prices have oscillated between $90 and $110 per barrel for most of the conflict so far, between 30 and 60 percent above preconflict levels. European LNG reached in late September $27 per million British thermal units (MMBtu) for European LNG, almost three times its level before the war.

In episodes like this, policymakers naturally focus on the supply loss. But fears of supply disruptions can also increase energy prices. Indeed, in January and February, before the start of the war, oil prices rose by about 10 percent amid rumors of a U.S. attack on Iran. Furthermore, shocks rarely occur in isolation. For instance, amid geopolitical tensions, exporting infrastructure may be hit while simultaneously large energy importers might decide to increase their purchases. Understanding the drivers of energy price movements is critical to calibrate policy appropriately.

When driven by stronger demand, higher energy prices reinforce broader inflationary pressures and generally support monetary tightening (Blanchard and Galí, 2008). When caused by actual supply disruptions, policymakers face a trade-off between containing inflation and avoiding further losses in economic activity (Woodford, 2003; Bandera et al., 2023), and a lot depends on the persistence of the shock and how it spills over to inflation expectations and broader price-setting (Natal, 2012). The policy implications are most uncertain when prices rise because of geopolitical risks, as inflationary effects could be temporary (Caldara et al., 2026).

This paper provides a framework to disentangle the drivers of energy prices in real-time, separately identifying, alongside traditional demand shocks, actual supply disruptions from geopolitical-risk shocks that increase the expectation of a future supply shortfall. The model builds on Gazzani et al. (2024) and, like them, our framework provides policymakers, and in particular central banks, with an operational tool to assess energy price movements as events unfold. Our main contributions are to propose a model that can distinguish between realized supply disruptions and geopolitical supply risk (or GSR) —-the possibility of future

disruptions— in real time, and to extend the framework to natural gas markets.

These two innovations are relevant because, as we will see, oil supply shocks have larger and more persistent effects on prices and unemployment than the risk of a future supply loss driven by geopolitical tensions, in line with the model in Kilian et al. (2024) that finds that GSR is not a key driver of global macroeconomic fluctuations. The framework is extended to analyze gas price fluctuations, a first in real-time, due to the growing importance of this energy source in Europe and the challenge fluctuations in gas prices represent for policymakers. The gas high-frequency decomposition complements earlier work at lower frequencies (Nick and Thoenes, 2014; Adolfsen et al., 2024).

The paper presents a new high-frequency structural vector autoregression (SVAR) model that uses daily energy prices, equity returns, and a newspaper-based measure of geopolitical risk from Caldara and Iacoviello (2022) to decompose energy prices into four shocks. Two of these shocks originate on the demand side, capturing changes in current economic conditions and revisions to expectations about future activity. The remaining two are supply-related, distinguishing realized disruptions to energy production from increases in geopolitical tensions that raise concerns about future supply shortfalls.

The application of the high-frequency SVAR tool to recent geopolitical conflicts —the 2022 Russian invasion of Ukraine and the ongoing War in the Middle East— and to April 2025 when the U.S. announcement of "Liberation Day" tariffs coincided with OPEC+'s announcement to increase production delivers some key takeaways. First, consistent with observational evidence, geopolitical supply risk exerts upward pressure on prices before supply disruptions materialize. Second, after supply disruptions materialize, heightened geopolitical risk continues to exert upward pressure on energy prices during conflicts. Thus, amid supply disruptions, part of the energy price increases reflect elevated risk premiums associated with uncertainty about future further disruptions, with widely different implications for policy given the distinct impacts on inflation and unemployment. Finally, during early April 2025, the announcement of tariffs, and its effect on dampening expectations of future energy demand, was the main culprit of the oil price drop at the time.

The shocks identified using the high-frequency SVAR are then used in local projections à la Jordà (2005) to estimate the impact of the different types of energy shocks on energy markets and the macroeconomy. In particular, we find that, as expected, oil supply shocks

decrease oil inventories, whereas increased risk of a supply shortfall increases them, in line with analyses by Kilian and Murphy (2014) or Kilian, Plante, and Richter (2024) that highlight the role of precautionary inventory demand amid elevated geopolitical tensions. Similarly, European gas storage increases, albeit with a delay, in response to a GSR shock. Furthermore, global oil production decreases on impact after a supply shock, but remains unchanged on impact after a shock to expectations about future production. Taken together, these results suggest that the high-frequency SVAR is able to effectively disentangle supply shocks from mere expectations of potential future shortfalls.

Turning now to the macroeconomic effects of the energy price shocks, our analysis uses data from 11 advanced and emerging oil-importing countries. The key and novel finding is that whereas supply oil shocks increase the aggregate price and the unemployment rate significantly, the effects of a GSR shock are more short-lived for prices, as they return to their preshock levels within six months to a year after the shock, and muted for unemployment. For gas, in a sample that includes only European countries, all shocks, including GSR shocks, have persistent effects on aggregate prices, and relatively muted impact on unemployment. The greater persistence of prices in response to gas shocks relative to oil shocks stems, at least partly, from the greater persistence that gas prices exhibit after any given gas shock.

The rest of the paper is organized as follows. Section 2 lays out the data, identification strategies, and estimation methodology for the oil and gas SVARs. Section 3 reports the transmission of energy shocks to financial and energy markets, and the macroeconomy more broadly. Section 4 provides a historical decomposition of recent large fluctuations in oil and gas prices using the high-frequency SVAR model. Of particular interest are large price increases during the recent wars in Ukraine and the Middle East and the imposition of new tariffs on "Liberation Day". Section 5 concludes.

2 Model, data, identification of shocks, and estimation

2.1 Overview of models: Shocks

The empirical models that we estimate use high-frequency (daily) financial and newspaper-based data, not subject to revision, to disentangle energy (oil and natural gas) prices into four

shocks: current demand, forward-looking demand, current supply, and geopolitically-driven supply disruption risk, henceforth geopolitical supply risk (GSR).

The decompositions of energy prices that this paper develops build on the work of Gazzani, Venditti, and Veronese (2024) and offers a real-time tool to decompose energy prices. Our framework introduces a fourth shock that allows to disentangle real supply disruptions from increases in oil prices that reflect higher geopolitical risk premia. We also extend the analysis to the European natural gas market (Title Transfer Facility, TTF).

In the remaining of this section, we describe each of the four shocks and provide examples to fix ideas.

The forward-looking demand shock captures unexpected changes to expectations about future energy demand. For instance, the enactment of trade tariffs that are likely to depress future world output growth and therefore future energy demand. Another example relates to the period of the COVID-19 pandemic, when news about the development of an effective vaccine increased expected future energy consumption. On November 9th, 2020, Brent price jumped by 7.2%, on news of clinical trials showing an effectiveness above 90%, even when lockdowns were still in place and current energy consumption was low.

Current demand shocks are typically characterized by cyclical changes to economic activity that affect energy demand, such as a slowdown in manufacturing, or for natural gas, weather anomalies (heatwaves or cold spells) that increase the demand for cooling or heating.

Current supply shocks include events that tighten global energy supply such as OPEC+'s announcements of changes in production or events such as wars or attacks that affect oil or gas supplies. Examples of the latter events abound: Iran-Iraq War, the Gulf War, or the Iraq War to name just a few. The most recent example is the the 2026 Middle East War that entailed the closure of the Strait of Hormuz, locking in the Gulf about 20 percent of world oil and liquefied natural gas (LNG) consumption.

The GSR shock refers to unanticipated increases in the probability of a future energy production shortfall amid elevated geopolitical risk (Kilian, Plante, and Richter, 2024). It refers to fears of future disruptions. Precautionary demand and stockpiling increase, as market participants fear future shortages. A notable example of a pure GSR was the Iran-Israel Twelve-Day War in June 2025, where oil and gas prices increased on fears of future disruptions that ultimately did not materialize. Similarly, for oil, Russia's invasion of Ukraine in 2022 was

also mostly a GSR shock as oil ——unlike gas— disruptions were minimal.

Table 1: Conceptual Decomposition of Energy Market Shocks

Shock	**Economic Interpretation**
Current demand	Physical demand conditions and contemporaneous economic activity.
Forward-looking demand	Expectations regarding future economic activity and energy demand.
Current supply	Realized supply shifts or disruptions affecting current market fundamentals.
Geopolitical supply risk	Expectations of future supply disruptions stemming from geopolitical events and tensions.

The next subsections describe in more detail the data used in both SVARs, the sample period, and the identification strategies in each energy market.

2.2 Oil model

The oil SVAR includes the following four variables: one-month Brent price, constructed from dated Brent, the price of a barrel of physical oil to be loaded within 10 and 30 days, and front-month Brent prices (or the most liquid futures contract traded), for which the maturity fluctuates from one to two months. The Appendix describes in detail how these two prices are combined to ensure that the oil price included in the SVAR refers to a contract whose maturity is always one month. The SVAR also includes a 13-month Brent price to capture future oil prices, and constructed by combining the 11-month and 13-month Brent futures price, such that price of the resulting synthetic contract always refers to barrel to be delivered 13 months forward. To capture global stock prices, we use the daily returns cumulated over a rolling one-year window of the MSCI ACWI Index, which covers large and mid-cap equities across 23 developed markets and 24 emerging markets countries, covering around 85% of the market capitalization in each country. Finally, the SVAR includes the 7-day moving average of the newspaper-based Geopolitical Risk Threat (GPRT) from Caldara and Iacoviello (2022), which measures the share of articles from ten U.S. newspapers that include keywords related to war threats, imperiled peace agreements, military buildups, nuclear threats, and terror threats, and is a subindex of the overall Caldara-Iacoviello geopolitical risk (GPR) index. The sample period in the analysis is between June 1, 2007, and May 29, 2026.

The SVAR model for oil can be written in compact form as:

$$A_0 y_t = A_+ x_t + e_t, \tag{1}$$

where $A_+ = \begin{bmatrix} A_1 & \cdots & A_p & c \end{bmatrix}$, p is the number of lags, and $x_t = \begin{bmatrix} y'_{t-1} & \cdots & y'_{t-p} & 1 \end{bmatrix}'$.

The accompanying reduced-form representation can be written as:

$$y_t = A_0^{-1} c + B_1 y_{t-1} + \cdots + B_p y_{t-p} + u_t, \qquad u_t \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \Sigma). \tag{2}$$

$B_k = A_0^{-1} A_k$ for $k = 1, \ldots, p$, and $u_t = A_0^{-1} e_t = D e_t$ and $\Sigma = (A_0' A_0)^{-1}$.

Matrix D is the key object in the structural identification of the model's four shocks: *(i)* forward-looking demand, *(ii)* current demand, *(iii)* current supply, and *(iv)* GSR. The identifying assumptions for each shock are summarized in Table 2. In what follows, we also provide support for the identification strategy followed.

Table 2: Summary of identifying restrictions imposed for oil.

	Future Demand	**Current Demand**	**Supply**	**GSR**
Oil one-month	IV	+ S	+	+
Oil 1-year-forward	IV	+ W		
Global equities	IV	+	-	
GPR Threats index				+

S indicates a relatively stronger magnitude response for front-month as compared to a relatively weaker magnitude response for 1-year-forward. + 90th indicates an absolute magnitude response above the 90th percentile.

(i) Forward-looking demand: It captures the link between oil as an asset class and other risky assets, with a flight to and from risky to safe assets during periods of very high or very low uncertainty. It is identified a proxy instrumental variable (IV) approach. Following Gazzani, Venditti, and Veronese (2024), if on a given day, equity volatility measured by the VIX, oil volatility measured by the OVX, and gold price changes are above the 90th percentile (below the 10th percentile) and oil front month prices changes are below the 10th percentile (above the 90th percentile), then the IV takes the value of the VIX change

for that given day. Otherwise, it takes the value of 0. Table 4 in the Appendix lists the days for which the IV is different from zero, which covers the likes of the COVID-19 pandemic, trade tensions, concerns about global growth, but also global growth rebounds and other positive events. The drivers of these events are varied, including geopolitical (e.g., trade tensions), but since the policy response does not depend on the underlying shock, we include them all under the forward-looking demand category.

(ii) *Demand:* It refers to a change in the current quantity of oil demanded and it is identified using sign and relative magnitude restrictions. For a positive demand shock, both oil one-month and 1-year-forward prices on a given day have to increase, along with an increase in the world equity index. Furthermore, the increase on the front end of the curve has to be larger in magnitude than that of the prices 1-year-forward, implying movement towards a more backwarded term structure. It also imposes a correlation between prices and equities, as higher oil prices due to strong economic activity tend to increase firms' cash flows (Kilian and Park, 2009; Ready, 2018).

(iii) *Supply:* Similarly to demand, it refers to a current change in quantity supplied and it is identified using sign restrictions. For a positive supply shock, one-month oil prices have to increase along with a decrease in world equity index, implying a negative co-movement between both variables. Intuitively, higher oil prices due to a supply shock hurt consumers' purchasing power, firms' revenues, and consequently equity returns (Kilian and Park, 2009; Ready, 2018).

(iv) *GSR:* Unanticipated increases in the risk of future oil supply disruptions increases energy prices due to increased precautionary inventory demand and higher risk premia (Alquist and Kilian, 2010; Kilian and Murphy, 2014), although current production remains unaffected. Thus, in our identification, we impose a positive comovement between one-month oil prices and the GPRT index. The inclusion of the GPRT index to identify geopolitical supply risk has several desirable properties. First, the Caldara-Iacoviello indices are widely used newspaper-based measures of geopolitical risk at daily frequency which spike around well-known geopolitical events, providing confidence that they capture the relevant episodes of geopolitical tension. Second, the Threats subindex of GPR captures heightened geopolitical risk (Caldara and Iacoviello, 2022; Verduzco-Bustos and Zanetti,

2026), while not explicitly including geopolitical acts such as beginning or escalation of war, that are in the GPR acts subindex and more likely to coincide with actual supply disruptions. Finally, the model will not assign to GSR, GPRT spikes with no impact on oil markets (e.g., terrorist threats in a non-oil producing country), as they will fail to increase front-month oil prices. Together, these features allow us to disentangle GSR shocks from actual oil supply disruptions, while disregarding geopolitical events with little relevance for oil markets.

2.3 Extending the framework to natural gas

The global natural gas market presents staggering differences compared to oil markets. Its chemical properties make the natural gas market difficult to be traded worldwide, as gas can be transported only through pipelines or needs to be liquefied (LNG) in order to be shipped overseas. Most countries rely on domestic consumption or import gas through inter-regional pipeline trade, as LNG requires building costly regasification import infrastructure, making the market for natural gas mainly segmented at regional levels.[1] Consequently, markets where LNG imports are limited remain largely unaffected by global spillovers. The North American natural gas market constitutes an example of this market type, with *Henry Hub* representing a benchmark for natural gas prices in the US. Conversely, other markets rely more on LNG imports, thus regional prices are directly affected by changes in global conditions. Examples of such markets include the European and the Northeast Asia markets, with the *Title Transfer Facility (TTF)* and the *LNG Northeast Asia* representing the benchmark prices for their respective regional gas markets.[2]

For these reasons, in this paper we use TTF prices as our benchmark for the natural gas price decomposition. Similarly to oil, the natural gas SVAR includes two price series: current TTF prices and future TTF prices, built, respectively, from front-month TTF futures and 13-month ahead TTF futures.[3] Natural gas prices display both roll jumps, associated with futures calendar changes, and seasonal fluctuations, stemming from the fact that winter

[1] According to the Energy Institute Statistical Review, total inter-regional trade accounted for roughly 898.7 billion cubic meters (bcm) in 2025, of which 403.5 bcm was pipeline trade and 495.2 bcm was LNG trade.

[2] In 2025, European (Asian) natural gas consumption amounted to 482.1 (976.1) bcm, of which 174 (360.3) bcm imported as LNG. In the same year, North American consumption was 1145.4 bcm, of which only 1.8 bcm from LNG imports.

[3] Front-month (13-month) contracts refer to time t prices for deliveries that occur at month $t+1$ ($t+13$).

demand is typically higher than summer demand and stockpile capacity is largely insufficient to completely smooth seasonal demand factors.[4] Therefore, we transform the TTF price series in order to control for these criticalities, and we provide further details in the Appendix. Also, comparably with oil, the natural gas SVAR includes European stock returns, computed as the cumulated daily returns over a one-year rolling window of the MSCI Europe Index. Finally, the SVAR includes the same GPRT measure employed in the oil analysis. The sample period runs from October 4, 2014 to May 29, 2026.

As for oil, the natural gas SVAR identifies the same for structural shocks, i.e. current demand, forward-looking demand, current supply and GSR. Table 3 summarizes the identifying restrictions. Similarly to oil, we identify current supply shocks and GSR through sign restrictions. For the former, we postulate that positive supply shocks produce increments in front-month TTF prices as well as a decrease in European stock valuations. For the latter, unanticipated rises in supply risk generate positive impacts on front-month TTF prices on impact; we also impose a positive comovement between front-month TTF prices and the GPRT index. No further restrictions are imposed on supply and GSR shocks. Differently from oil, we identify forward-looking demand through sign restrictions and current demand shocks using a proxy IV approach. In the SVAR we assume that a positive forward-looking demand shock generates a positive response of 1-year forward prices on impact, along with an increase in the European equity index. As natural gas markets are less financialized than oil, current demand is strongly influenced by weather conditions. During the cold season, when demand for heating is higher, unexpected cold spells can temporarily increase demand for natural gas. In a similar fashion, during the summer unexpected heatwaves can increase gas-fueled electricity generation for cooling purposes (see Mu, 2007 and Saha et al., 2014). Hence, we employ daily forecasted temperature anomalies, expressed in Total Degree Days (TDDs) as a proxy for current demand. On the one side, in the SVAR higher forecasted TDDs indicate either forecasts of cold spells or heatwaves, which may lead to higher-than-expected demand for natural gas. On the other side, lower forecasted TDDs indicate temperatures close to the historical averages for the period, when demand aligned with expectations. In the Appendix we provide further details on the construction of our weather proxy for current demand.

[4] In the EU, the technical storing capacity for natural gas amounts to 1131.57 TWh, according to Gas Infrastructure Europe (GIE). As annual natural gas consumption in 2025 amounted to 3519 TWh (see Energy Institute, 2026), stockpiles can cover at most 117.37 days of annual consumption.

Table 3: Summary of identifying restrictions imposed for natural gas.

	Future Demand	**Current Demand**	**Supply**	**GSR**
TTF one-month		IV	+	+
TTF 1-year-forward	+	IV		
European equities	+	IV	-	
GPR Threats index		IV		+

2.4 Estimation

Both models are estimated using Bayesian methods within the European Central Bank's (ECB) BEAR toolbox (Dieppe et al., 2016), setting the number of lags, p, to 4. This subsection gives an overview of the main estimation details and Section A.2 in the Appendix offers the technical details, including priors' functional forms and parametrization.

Identification by a combination of proxy IV and sign restrictions implies that the reduced-form representation of the VAR in equation (2) is augmented with a measurement equation that links the proxy IV m_t, the variable of events for oil or the weather anomalies for TTF, with the corresponding structural shock, forward demand for oil and current demand for TTF. In particular, the measurement equation takes the following form:

$$m_t = \beta e_{FD(CD),t} + v_t \qquad v_t \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1). \tag{3}$$

where e_{FD} is the structural shock corresponding to forward demand for oil and to the structural shock to current demand (CD) in the case of natural gas. v_t is a measurement error orthogonal to all structural shocks of the VAR model, e_t.

The BEAR toolbox applies the algorithm in Caldara and Herbst (2019), by which the proxy IV pins down column $b_{FD(CD)}$, the elements of the reduced-form matrix that correspond to the forward demand (current demand) shock for oil (natural gas), and the sign and magnitude restrictions enter in a second step inside the algorithm. BEAR then checks the responses given by each iteration. If the sign and relative magnitude restrictions are respected, it keeps the draw, otherwise if rejects it. We simulate 2000 iterations, with the first 200 used as

burn-in.

3 Transmission of shocks

This section investigates the economic transmission of the four shocks identified in the high-frequency SVARs. We begin by documenting their dynamic effects on financial markets using the models' impulse response functions. We then validate the identification strategy by examining the responses of oil supply and energy inventories, which provide useful benchmarks for distinguishing among the different shocks, using local projections (LPs). Finally, using LPs too, we analyze the implications of these shocks for policy-relevant macroeconomic outcomes (the aggregate price level and unemployment).

3.1 Financial variables impulse response functions

The oil SVAR's impulse response functions, in panel (a) of Figure 1, show that both forward-looking and current demand increase oil prices —one-month and futures— and equities. However, while current demand promotes backwardation (bigger increase of front-month than futures) in line with the temporary nature of the shock, forward-looking demand causes an upward shift of the futures curve. Similarly to current demand, a current supply shock promotes backwardation, as physical market tightens —an assumption that was not imposed in the identification strategy, but that the model nevertheless delivers. As imposed by the sign restriction, equities fall after a supply shock, consistent with lower firms' profits.

The results for gas, shown in panel (b) of Figure 1, exhibit some important differences. First, current demand shocks do not affect European equities, as current demand is primarily driven by weather conditions, which have little bearing on the broader European stock market. Second, the impact of future demand shocks on front-month TTF prices is not statistically significant, consistent with weather remaining the dominant driver of short-term fluctuations in gas prices. Finally, the response of gas prices (front-month and futures) to current demand (and GSR) is more persistent than for oil.

Turning to the GSR shock, the main difference with the actual supply disruption concerns the response of equities. While for both gas and oil a supply shock leads to a fall in equity

prices, their response to the GSR shock is not significantly different from zero. Although previous work has found a link between the keyword-based GPRT and the stock market (Gonçalves et al., 2025); the link is often weak, especially when focusing on stock returns (International Monetary Fund, 2025). According to Sheng et al. (2025), stronger equity responses hinge on the inclusion of trade tensions in geopolitical risk measures. However, these tensions are not in the GSR shock, but in the forward-looking demand shock (see Table 4 in the Appendix), because they depress aggregate demand and oil prices.

3.2 Impact on energy markets

To validate the high-frequency SVARs' shocks, daily shocks are aggregated to monthly frequency and their impact on global oil production, oil inventories, and natural gas inventories are analyzed using local projections (LPs) (Jordà, 2005).[5]

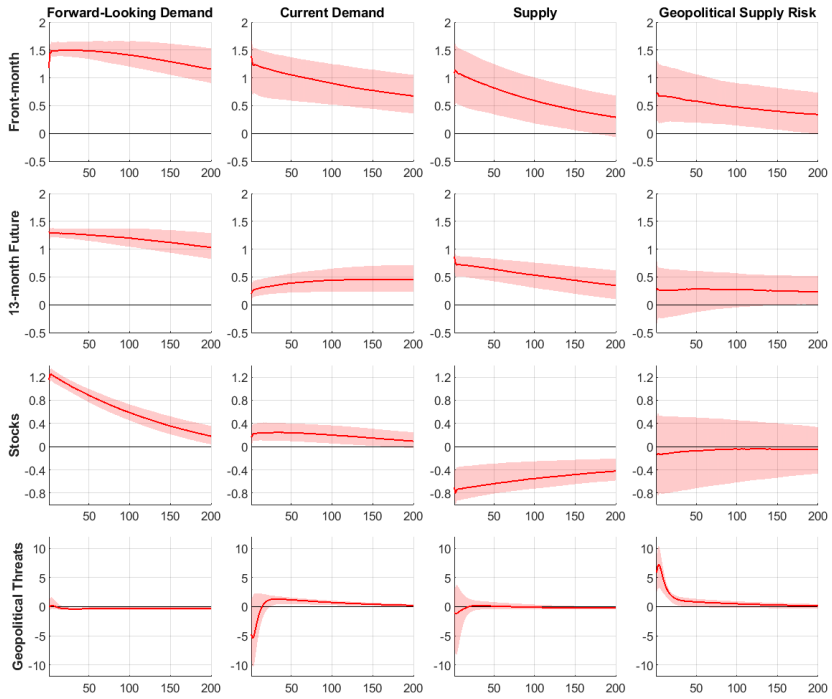
We run the following LP equation:

$$y_{t+h} - y_{t-1} = \alpha_h + \beta_h Shock_t + \theta_h Covid_t + \delta_h \left(Shock_t \times Covid_t\right) + \sum_{l=1}^{12} \phi_{l,h} \Delta y_{t-l} + \varepsilon_{t+h} \quad (4)$$

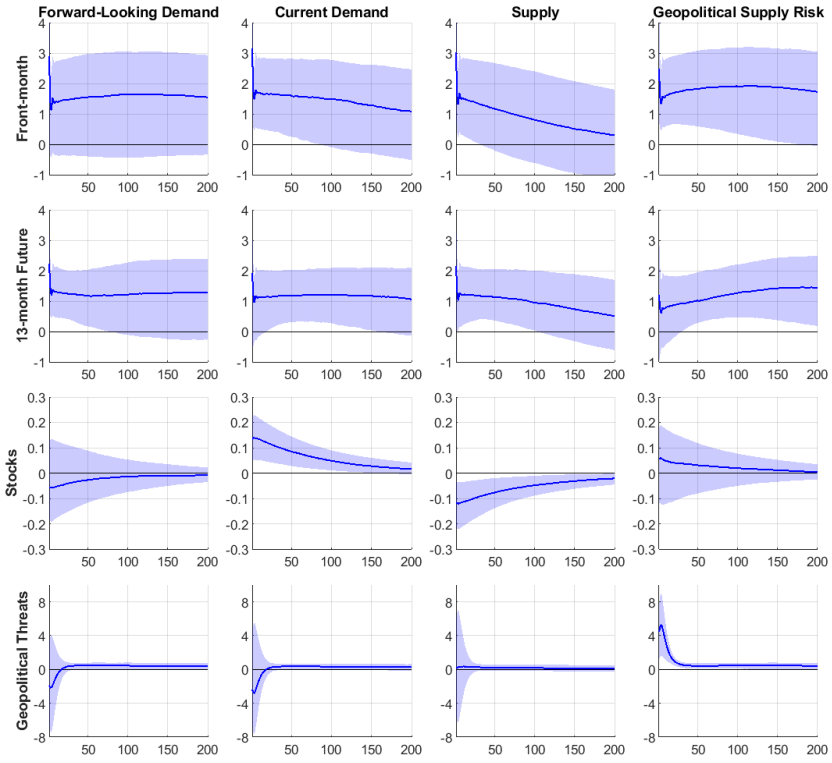
where y_t is the relevant global energy market indicator and $Shock_t$ the relevant shock identified from the high-frequency VAR. $Covid_t$ is a dummy variable equal to 1 for observations during March-June 2020, to account for how the pandemic might have affected the usual relationship between variables. Outside of the Covid-19 window, β_h is the response of y_t to $Shock_t$ at horizon h, whereas during this window the response is $\beta_h + \delta_h$. We include 12 lags of the endogenous variable y_t as controls (Jordà, 2023). The local projections are estimated over the sample 2007:m6 - 2026:m5 for oil, and 2014:10 - 2026:m5 for gas. The LP for gas inventories takes into account the change in European gas storage regulation of 2022, and adds another dummy $Regulation_t$ (and its iteractions), that equals 1 from July 2022 onwards.

As shown in Figure 2, forward-Looking demand, current demand and supply shocks have a negative impact on oil inventories. The impact of forward-looking demand shocks appears to be temporary, with inventories returning to their initial levels within 2 years, after

[5]The Appendix lists the data sources for these series. Lack of data for natural gas production at the monthly (and quarterly) frequency prevents us from studying the response of natural gas production to the gas SVAR shocks.



(a) Oil SVAR impulse response functions



(b) Natural gas SVAR impulse response functions

Figure 1: Responses of the variables in SVAR models to the shocks

Notes: These figures plot the impulse response functions of the models' variables (in rows) to the models' shocks (in columns). Shocks are a one-standard-deviation shock. The solid lines are the median responses and the shaded area corresponds to the 68% confidence interval. Panel (a) plots the responses for the oil SVAR and panel (b) the responses for the natural gas SVAR.

a peak drawdown of 0.1% for a shock that increases one-month prices by 1%. Current demand and supply shocks tend to have a more persistent impact on inventories, with a drawdown of about 0.2% after 2 years. Conversely, a geopolitical supply risk shock has a positive impact on inventories, consistent with a precautionary inventories buildup, as shown theoretically by Kilian, Plante, and Richter (2024).[6] Our empirical results suggest that the high-frequency VAR is able to disentangle actual oil supply disruptions from pure geopolitical supply risks shocks.

The results for gas inventories are broadly similar to the ones for oil and are in Figure 7 in the Appendix. Demand-side gas shocks decrease European gas inventories, whereas GSR gas shocks increases them, although with a delay. The decrease of gas storage amid gas supply shocks occurs on impact but it is less pronounced and more short-lived than for oil.

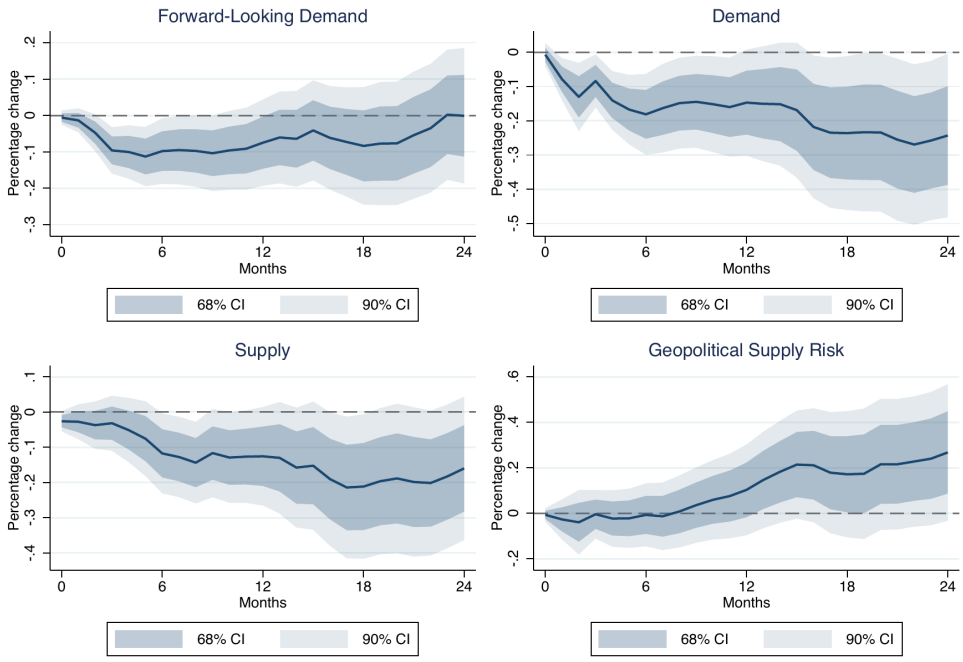


Figure 2: OECD inventories responses to oil shocks.

Notes: Responses to shocks normalized to raise the front-month oil price by 1 percent on impact. The shaded area corresponds to the 68-90% confidence interval.

Looking at global oil production in Figure 3 forward-looking demand shocks appear to have a delayed and a persistent impact, consistent with a shift in expected future oil demand. Current demand shocks do not appear to have a significant effect on global oil production.

[6]The results in Figure 2 use data over 2006-2026 on OECD inventories to approximate global inventories. Using an alternative measure of global inventories that covers non-OECD countries between 2017 and 2026, does not change the signs of the median responses. However, the responses become noisier given the shorter sample. See Figure 8 in the Appendix.

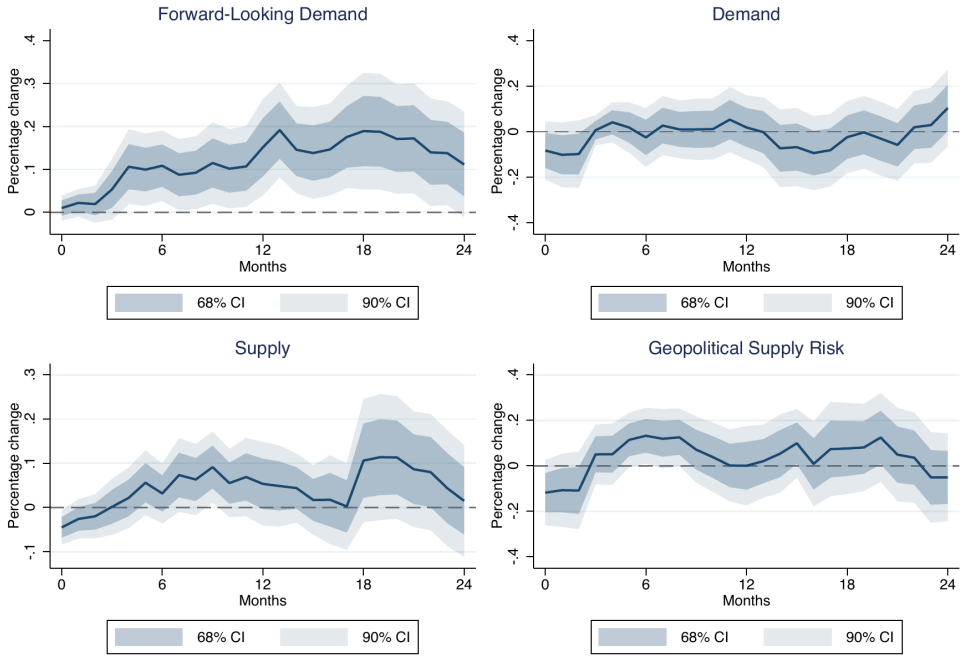


Figure 3: Global oil production responses to oil shocks.

Notes: Responses to shocks normalized to raise the front-month oil price by 1 percent on impact. The shaded area corresponds to the 68-90% confidence interval.

Supply shocks have a contemporaneous negative effect on global oil production, which then rises temporarily over its initial level. Finally, global oil production does not exhibit a significant change on impact after a GSR shock.

3.3 Macroeconomic effects: consumer prices and unemployment

To assess the macroeconomic consequences of the identified shocks, we estimate monthly panel LPs for consumer prices and unemployment. The analysis focuses on whether GSR shocks transmit differently from actual supply disruptions. Following a similar approach to the energy market exercise, we aggregate the daily shocks to monthly frequency. The oil analysis covers eleven oil-importing advanced and emerging market economies: Germany, Spain, France, Indonesia, Italy, Japan, South Korea, Turkey, the United Kingdom, the United States, and South Africa. The LNG analysis focuses on five European countries—Germany, France, Italy, Spain, and the United Kingdom—reflecting the regional relevance of TTF prices.[7] Each shock is normalized to raise the relevant commodity price at the short-end of the curve

[7]The Appendix lists the data sources for these macroeconomic aggregates.

(one-month for oil or seasonally adjusted front-month for gas) by 1 percent on impact.

For CPI, the dependent variable is the cumulative log change in the price level from the pre-shock month $t-1$ to horizon $t+h$. For each horizon, we estimate

$$100\left[\log(CPI_{i,t+h}) - \log(CPI_{i,t-1})\right] = \alpha_{i,h} + \beta_h Shock_t + \theta_h Covid_t + \delta_h(Shock_t \times Covid_t) + \sum_{l=1}^{12} \phi_{l,h}\pi_{i,t-l} + \varepsilon_{i,t+h}, \tag{5}$$

where $\alpha_{i,h}$ denotes country fixed effects, $Shock_t$ is one of the SVAR-identified oil or LNG shocks, and $\pi_{i,t} = 100\Delta\log(CPI_{i,t})$ is monthly CPI inflation.

For unemployment, the dependent variable is the unemployment rate in levels at horizon $t+h$:

$$u_{i,t+h} = \alpha_{i,h} + \beta_h Shock_t + \theta_h Covid_t + \delta_h(Shock_t \times Covid_t) + \sum_{l=1}^{12} \phi_{l,h} u_{i,t-l} + \varepsilon_{i,t+h}. \tag{6}$$

The controls therefore comprise of twelve lags of monthly inflation in the CPI specification and twelve lags of the unemployment rate in the unemployment specification. Country fixed effects absorb time-invariant cross-country heterogeneity, and standard errors are clustered at the country level. We estimate a separate panel regression at each horizon $h = 0, \ldots, 24$, entering each of the four shock types separately for oil and LNG.

Both specifications include a COVID-19 dummy equal to one from March through June 2020, together with its interaction with the shock. These terms allow both the conditional mean of the outcome and the transmission of commodity-market shocks to differ during the onset of the pandemic. Outside this window, β_h measures the response at horizon h; for shocks occurring during the COVID-19 window, the corresponding response is $\beta_h + \delta_h$. The macroeconomic LNG specifications do not include the July-2022 regulation interaction used in the LNG market exercise.

The sequence $\{\beta_h\}_{h=0}^{24}$ traces the baseline response over two years. For CPI, the coefficients measure the effect of the shock on the cumulative percentage change in the price level between the pre-shock month $t-1$ and horizon $t+h$. For unemployment, the coefficients measure percentage-point effects on the unemployment rate relative to its counterfactual path, conditional on the included controls.

The oil results show positive price-level responses to all four shocks, although their magnitude and persistence differ considerably (panel (a) of Figure 4). Forward-looking demand, current demand, and actual supply shocks normalized to raise oil prices by 1 percent increase the CPI level by approximately 0.04–0.10 percent after one year. This range is broadly in line with other estimates found in the literature (Choi et al., 2018; Känzig, 2021), though slightly larger. In contrast, the initial increase in the price level following oil GSR shocks dissipates within approximately six months. The short-lived impact of the GSR shock on inflation echoes the finding in Caldara et al. (2026) when studying geopolitical risk and inflation and is consistent with the model in Ambrosino et al. (2026). Supplementary estimates for core CPI reinforce this distinction: actual supply disruptions generate a gradual and persistent increase in core prices, whereas GSR shocks produce a smaller initial increase than in headline CPI that subsequently dissipates. These results are consistent with more persistent price pressures extending beyond the direct energy component following actual supply disruptions. Supplementary estimates for core CPI, reported in Appendix Figure 9, reinforce this distinction.

The unemployment responses also differ across shocks (panel (b) of Figure 4). Demand-driven oil shocks, particularly forward-looking demand shocks, lower unemployment relative to its counterfactual path, with the estimated response remaining negative for almost two years. An adverse supply shock normalized to raise oil prices by 1 percent increases unemployment by approximately 0.03 percentage points after one year. The unemployment response to oil GSR shocks is not statistically distinguishable from zero.

These findings suggest that distinguishing actual oil supply disruptions from GSR shocks matters for assessing their macroeconomic consequences. Actual supply disruptions generate more persistent estimated increases in the price level and unemployment, whereas oil GSR shocks produce a shorter-lived price-level response and no statistically discernible unemployment response. The exercise also highlights the value of separating the risk of supply disruptions from: (i) geopolitical shocks that increase macroeconomic uncertainty (for instance, terrorist attacks in non-energy exporting countries), which is included in the forward-looking demand shock, and (ii) actual supply disruptions since these two other shocks tend to have bigger impacts on economic activity (Kilian, Plante, and Richter, 2024; Verduzco-Bustos and Zanetti, 2026).

The European price-level responses to LNG shocks are more persistent than those to

their oil counterparts, particularly for GSR shocks (panel (a) of Figure 5) and are in line quantitatively with the passthrough of gas supply shocks to the price level found in López et al. (2026). For shocks normalized to raise TTF prices by 1 percent, all four shock types generate sustained increases in the CPI level. The price-level responses to LNG shocks, particularly GSR shocks, are more persistent than those to their oil counterparts. Whereas the estimated response to oil GSR shocks dissipates relatively quickly, the response to LNG GSR shocks remains statistically distinguishable from zero for considerably longer. This pattern is consistent with the more persistent commodity-price responses to LNG shocks, especially GSR shocks, documented in the daily-frequency IRFs in section 3.1 and shown in Figure 1.

The unemployment responses to LNG shocks are comparatively muted (panel (b) of Figure 5). Demand shocks are associated with modest declines in unemployment, while the responses to actual supply disruptions and GSR shocks are not statistically distinguishable from zero. The absence of a statistically significant unemployment response to GSR shocks is consistent with the oil results. For supply shocks, Moll et al. (2024) highlights the power of substitution ——geographical and of energy sources— in explaining very mild recessions after natural gas supply shock. More broadly, one possible explanation for the weaker unemployment responses to LNG shocks is the difference in energy end uses: natural gas plays an important role in heating and cooling, whereas oil is heavily used in transportation (International Energy Agency, 2025). However, the LP estimates do not separately identify this mechanism.

Taken together, the results underscore the importance of distinguishing actual supply disruptions from geopolitical supply risk when assessing the macroeconomic effects of energy shocks. This distinction is particularly clear for oil: actual supply disruptions generate persistent increases in headline and core prices and higher unemployment, whereas GSR shocks produce only short-lived positive price responses and no statistically significant unemployment response. While LNG GSR shocks generate more persistent price-level increases in Europe than oil GSR shocks, their unemployment effects also remain statistically insignificant. Thus, supply risk and actual disruptions have distinct macroeconomic implications, with the commodity market shaping how those differences manifest.



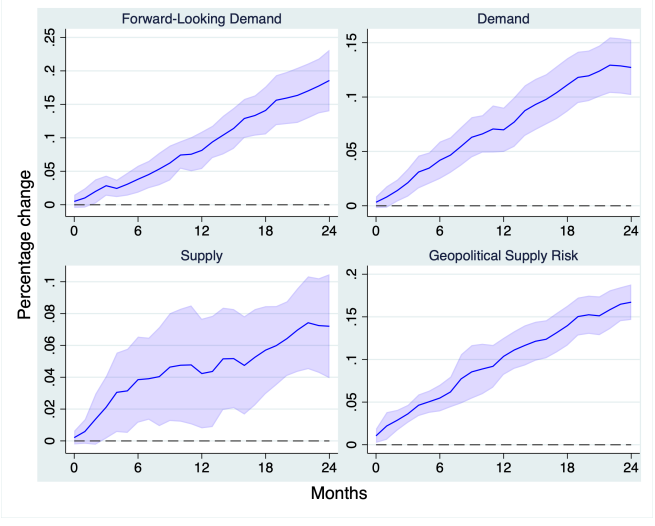
(a) Consumer prices



(b) Unemployment

Figure 4: Impulse responses of consumer prices and unemployment to oil shocks.

Notes: Responses to shocks normalized to raise the front-month oil price by 1 percent on impact. Panel (a): cumulative CPI log changes from $t-1$, multiplied by 100. Panel (b): unemployment responses in percentage points. Solid lines show estimated responses outside the COVID-19 window; shaded areas denote 90 percent pointwise confidence intervals. Horizons are in months.



(a) Consumer prices



(b) Unemployment

Figure 5: Impulse responses of consumer prices and unemployment to LNG shocks.

Notes: Responses to shocks normalized to raise the front-month TTF price by 1 percent on impact. Panel (a): cumulative CPI log changes from $t-1$, multiplied by 100. Panel (b): unemployment responses in percentage points. Solid lines show estimated responses outside the COVID-19 window; shaded areas denote 90 percent pointwise confidence intervals. Horizons are in months.

4 Historical episodes

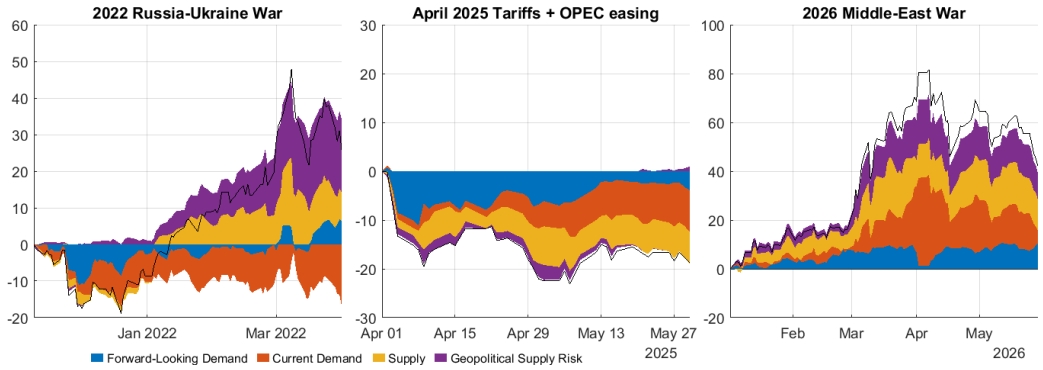
The analytical framework presented above is particularly useful in periods of multiple concurrent shocks and amid geopolitical escalations. To illustrate the models' ability to disentangle shocks in real time , front-month oil and gas prices are decomposed into their main drivers during three relevant and recent historical episodes: *(i)* the beginning of the Russia-Ukraine War in 2022, *(ii)* the U.S. announcement of tariffs in early April 2025, and *(iii)* the 2026 War in the Middle East.

The first and third episodes allow us to explore geopolitical conflicts with varying degrees of supply disruptions. The Russia-Ukraine War in 2022 entailed a major supply disruption for European gas markets, with TTF prices quadrupling in six months, whereas the oil disruption was minimal. Still, oil prices spiked. In contrast, both oil and gas supply were heavily disrupted during the 2026 War in the Middle East, and fears that the geopolitical situation would worsen were also paramount. During the second episode (early April 2025), energy markets faced two almost simultaneous surprise announcements: the U.S. announcement of broad new trade tariffs on April 2nd and the OPEC+'s surprise unwinding of oil production cuts on April 3rd. Consequently, oil and European gas prices dropped; but a key question arises: how much of the oil price drop was due to higher supply and how much was due to lower demand?

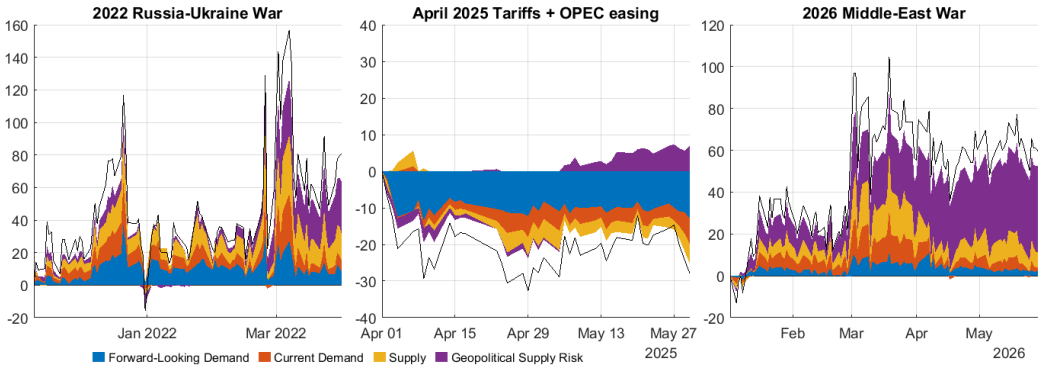
The early April 2025 oil decomposition (Figure 6, panel (a), middle chart) shows that forward-looking demand and supply shocks were the main culprits of the drop in oil prices. Given that tariffs tend to depress future energy demand, the decomposition favors the tariff announcement as the main driver of the initial oil price drop. The impact of OPEC+'s increased production plan (supply) was relatively minor initially. However, as details on tariffs emerged and exceptions were introduced, the contribution of forward-looking demand shrank. In May 2025, the contribution of supply rose on rumors of OPEC+ further unwinding production cuts. The drop in expected future energy demand is also apparent in the early April TTF price drop, as mostly forward-looking demand exerts downward pressure on gas prices (Figure 6, panel (b), middle chart). The decomposition of gas prices also shows that supply played no role in the early April price drop, lending support to the methodology, as OPEC+ is an association of oil producers and their decision in early April was about oil production.

Focusing now on the 2022 and 2026 conflicts (Figure 6, remaining charts), GSR exerted upward pressure on energy prices —both oil and gas— before the wars began , reflecting rising expectations of military escalation. Once the conflicts started, the decomposition of oil prices reveals notable differences across episodes. During the Russia-Ukraine war, the contribution of GSR was more than twice as large as that of actual supply, whereas during the 2026 Middle East war the two shocks contributed roughly equally. This difference is consistent with the greater severity of the actual supply losses in the latter episode. In contrast, supply emerges as a key driver of gas price movements during the Russia-Ukraine, reflecting the prominent gas supply disruptions in that conflict. In the 2026 Middle East war, GSR put substantial upward pressure on gas prices, followed by supply, again in line with the elevated uncertainty and shipping affected by the closure of the Strait of Hormuz and damages to Qatari facilities. The contribution of demand, however, is similar across oil and gas during the 2026 war. It exerts upward pressure on prices at the beginning of the war, as refineries and electrical companies likely competed for the little crude and LNG available. As the war continues, the decomposition starts showing a lower demand contribution, signaling demand compression. A key takeaway from these decompositions is that realized energy shortages do not eliminate the influence of geopolitical risk. Even after supply disruptions materialize, heightened GSR continues to exert upward pressure on energy prices during these episodes. Hence, not all of the observed increase in energy prices during geopolitical crises should be interpreted as evidence of physical scarcity; part of the rise reflects elevated risk premia associated with uncertainty about future disruptions.

To be clear, these decompositions are in quasi-real time. Figure 10 in the Appendix show the decompositions in real time. The decompositions for the two wars and both energy sources are very similar in real time to the ones presented in this section. The gas decomposition for the April 2025 leaves a slightly larger portion unexplained, but still assigns to forward-looking demand the majority of the drop and assigns no contribution of supply in the LNG price drop. For oil, the supply contribution shrinks slightly in favor of forward-looking demand during the early April 2025 episode.



(a) Oil one-month price decompositions



(b) Natural gas front-month decompositions

Figure 6: Variance decompositions of front-month energy prices during three episodes

Notes: Variance decompositions of front-month prices energy prices using the four-variable SVAR models. Units in the horizontal axis are percent deviation from the starting point at the beginning of the period considered. Data used to estimate the models are June 1, 2007-May 29, 2026, for oil; October 4, 2013-May 29, 2026 for natural gas, daily frequency.

5 Conclusions

Effective policy responses to energy-related shocks, especially monetary policy, require identifying the source of energy price movements in real time. Because the macroeconomic implications are different, central bankers should be able to distinguish between price increases driven by demand shocks and those driven by supply shocks and, among the latter, between price increases due to geopolitical tensions and those caused by actual supply disruptions. For example, whereas oil-related geopolitical supply risk typically has only short-lived effects on prices and no measurable effect on unemployment, supply disruptions and forward-looking demand shocks generate larger and more persistent macroeconomic effects.

Policymakers' task is further complicated by the fact that shocks usually arise simultaneously. Recent events in the Middle East show that energy prices often increase in anticipation of supply disruptions and that geopolitical risk continues to influence prices even when shortages have already occurred. As a result, interpreting higher energy prices solely as evidence of physical scarcity can lead to inappropriate policy responses.

The new high-frequency structural model proposed in this paper allows to disentangle between energy price increases driven by shortages or fear of them for both oil and natural gas prices. This tool can help policymakers calibrate policy appropriately, thereby supporting more timely and effective policy responses. Whereas the focus of this paper is on oil and gas, the framework is flexible enough to be extended to other commodities. This extension is left for future research.

References

Adolfsen, Jakob K., Marco Ferrari Minesso, Knut Einar Mork, and Ine van Robays (2024). "Gas Price Shocks and Euro Area Inflation". *Journal of International Money and Finance* 149, p. 103183.

Alquist, Ron and Lutz Kilian (2010). "What Do We Learn from the Price of Crude Oil Futures?" *Journal of Applied Econometrics* 25.4, pp. 539–573.

Ambrosino, Ludovica, Jenny Chan, and Silvana Tenreyro (2026). "Trade fragmentation, inflationary pressures and monetary policy". *Journal of International Economics* 163. International Seminar on Macroeconomics, 2025, p. 104219.

Bandera, Natalia, Lucy Barnes, Mathias Chavaz, Silvana Tenreyro, and Lien von dem Berge (2023). "Monetary Policy in the Face of Supply Shocks: The Role of Inflation Expectations". *ECB Forum on Central Banking*. Sintra, Portugal.

Blanchard, Olivier J. and Jordi Galí (2008). *The Macroeconomic Effects of Oil Price Shocks: Why are the 2000s so different from the 1970s?* NBER Working Paper 13368. National Bureau of Economic Research.

Caldara, Dario, Sam Conlisk, Matteo Iacoviello, and Michael Penn (2026). "Do Geopolitical Risks Raise or Lower Inflation?" *Journal of International Economics* 159, p. 104188.

Caldara, Dario and Edward Herbst (2019). "Monetary Policy, Real Activity, and Credit Spreads: Evidence from Bayesian Proxy SVARs". *American Economic Journal: Macroeconomics* 11.1, pp. 157–192.

Caldara, Dario and Matteo Iacoviello (2022). "Measuring Geopolitical Risk". *American Economic Review* 112.4, pp. 1194–1225.

Choi, Sangyup, Davide Furceri, Prakash Loungani, Saurabh Mishra, and Marcos Poplawski-Ribeiro (2018). "Oil prices and inflation dynamics: Evidence from advanced and developing economies". *Journal of International Money and Finance* 82, pp. 71–96.

Dieppe, Alistair, Romain Legrand, and Björn Van Roye (2016). "The BEAR toolbox".

Energy Institute (2026). *2025 Statistical Review of World Energy, 74th edition.* Energy Institute.

Gazzani, Andrea, Fabrizio Venditti, and Giovanni Veronese (2024). "Oil price shocks in real time". *Journal of Monetary Economics* 144, p. 103547.

Gonçalves, Andrei S., Alessandro Melone, and Andrea Ricciardi (2025). *The Pricing of Geopolitical Tensions over a Century*. SSRN Working Paper No. 5218714.

International Energy Agency (2025). *World Energy Outlook 2025*. Paris: International Energy Agency.

International Monetary Fund (Apr. 2025). *Global Financial Stability Report: Chapter 2. Geopolitical Risks: Implications for Asset Prices and Financial Stability*. Tech. rep. Washington, DC: International Monetary Fund.

Jordà, Òscar (2005). "Estimation and Inference of Impulse Responses by Local Projections". *American Economic Review* 95.1, pp. 161–182.

Jordà, Òscar (2023). "Local Projections for Applied Economics". *Annual Review of Economics* 15.Volume 15, 2023, pp. 607–631.

Kilian, Lutz and Daniel P. Murphy (2014). "The Role of Inventories and Speculative Trading in the Global Market for Crude Oil". *Journal of Applied Econometrics* 29.3, pp. 454–478.

Kilian, Lutz and Cheolbeom Park (2009). "The Impact of Oil Price Shocks on the U.S. Stock Market". *International Economic Review* 50.4, pp. 1267–1287.

Kilian, Lutz, Michael D. Plante, and Alexander W. Richter (2024). *Geopolitical Oil Price Risk and Economic Fluctuations*. Working Papers 2403. Federal Reserve Bank of Dallas.

Känzig, Diego R. (2021). "The Macroeconomic Effects of Oil Supply News: Evidence from OPEC Announcements". *American Economic Review* 111.4, 1092–1125.

López, Lucia, Florens Odendahl, Susana Párraga Rodríguez, and Edgar Silgado-Gómez (2026). "The pass-through to inflation of gas price shocks". *Journal of Economic Dynamics and Control* 182, p. 105218.

Moll, Benjamin, Moritz Schularick, and Georg Zachmann (2024). "The Power of Substitution: The Great German Gas Debate in Retrospect". *Brookings Papers on Economic Activity* 2023.Fall, pp. 395–455.

Mu, Xiaoyi (2007). "Weather, storage, and natural gas price dynamics: Fundamentals and volatility". *Energy Economics* 29.1, pp. 46–63.

Natal, Jean-Marc (2012). "Monetary Policy Response to Oil Price Shocks". *Journal of Money, Credit and Banking* 44.1, pp. 53–101.

National Weather Service (2026). *Heating and Cooling Degree Days*. Available at: https://www.weather.gov/key/climate_heat_cool (accessed July 7, 2026).

Nick, Sebastian and Simone Thoenes (2014). "What Drives Natural Gas Prices? A Structural VAR Approach". *Energy Economics* 45, pp. 517–527.

Ready, Robert C (2018). "Oil prices and the stock market". *Review of Finance* 22.1, pp. 155–176.

Saha, Suranjana, Shrinivas Moorthi, Xingren Wu, Jiande Wang, Sughir Nadiga, Patrick Tripp, David Behringer, Yu-Tai Hou, Hui-ya Chuang, Mark Iredell, Michael Ek, Jesse Meng, Rongqian Yang, Malaquias Peña Mendez, Huug van den Dool, Qin Zhang, Wanqiu Wang, Mingyue Chen, and Emily Becker (2014). "The NCEP Climate Forecast System Version 2". *Journal of Climate* 27.6, pp. 2185–2208.

Sheng, Jian, Zhe Sun, and Qiang Wang (May 2025). "Geopolitical Risk and Stock Returns". Mimeo.

Sims, Kelly, Andrew Reith, Eddie Bright, Jason Kaufman, Joe Pyle, Justin Epting, Jack Gonzales, Daniel Adams, Eric Powell, Marie Urban, and Amy Rose (2023). *LandScan Global 2022 [Data set]*. Tech. rep. Oak Ridge National Laboratory.

Verduzco-Bustos, Gustavo and Francesco Zanetti (2026). "The Effects of Geopolitical Oil Price Shocks". Mimeo, April.

Woodford, Michael (2003). *Interest and Prices: Foundations of a Theory of Monetary Policy*. Princeton, NJ: Princeton University Press.

A Online Appendix

A.1 Data

A.1.1 Brent price adjustments

Depending on the day within a month, the expected delivery date of the Front-month Brent varies from 1 to 2 months. Indeed, Front-month Brent experiences recurring changes as the reference period changes each month, which could be sizable in periods of extreme backwardation. For example, as the reference period of the Front-month Brent moved from May to June on April 1st 2026, its price dropped by about \$17, amid markets expecting a rapid resolution of the Middle East War. Another issue relates to a change introduced in the expiry calendar of Brent contracts on January 29, 2016. As a result of this change, prices after this date refer to deliveries roughly one month further ahead compared to prices until then. To increase prices comparability across time, we compute the price of a synthetic barrel of oil for which the delivery date is always one month ahead. This is done by computing the weighted average of the Front-month Brent and the Dated Brent. The latter is the spot price of a physical barrel of oil to be delivered 10 to 30 days ahead. More formally, if p_t^{FM} and p_t^{Dated} are respectively the front-month and Dated Brent prices, we define the one-month Brent price as:

$$p_t^{1M} = \alpha_t \cdot p_t^{FM} + (1 - \alpha_t) \cdot p_t^{Dated}, \tag{7}$$

where the weights α_t are solutions of the following equation:

$$\alpha_t \cdot d_t^{FM} + (1 - \alpha_t) \cdot d_t^{Dated} = 30, \tag{8}$$

with d_t^{FM} the delivery date (in days) of Front-month Brent at day t, and $d_t^{Dated} = 20$ the delivery date (in days) of Dated Brent (median of the 10 to 30 days interval).

A.1.2 TTF price adjustments

In this paper we employ two prices series: front-month, i.e. 1-month ahead futures, and 13-month ahead futures. We use front-month futures and refer to them as current prices and, in a similar fashion, 13-month ahead futures are referred to as 1-year ahead prices. Taken at

face value, natural gas futures prices present two issues that need to be addressed: (i) jumps associated with futures calendar changes and, (ii) seasonal fluctuations. Each future contract is associated with physical deliveries that need to take place within a predetermined month. For instance, the November 2026 TTF futures contract refers to deliveries that take place between November 1st 2026 and November 30th 2026. On October 1st 2026, the November 2026 TTF futures contract becomes the front-month futures contract and replaces the September 2026 contract. Hence, October 1st 2026 is a calendar change date. On this date, price jumps may occur driven by the fact that the two different futures contracts reflect different delivery dates, where demand may be higher or lower.

Formally, for $t \in \{0, ..., T\}$ we define with $F_t^{c_t}$ the futures price at time t referred to the delivery contract c_t, and let $p_t = F_t^{c_t}$ be the front-month price of the period. Let $\mathcal{R}$ be the collection of all *roll dates*, i.e. the set of all dates in which contracts change the delivery date, which we define as $\mathcal{R} = \{\tau : c_\tau \neq c_{\tau-1}\}$. On a calendar change date, the front-month prices difference is given by:

$$p_\tau - p_{\tau-1} \equiv F_\tau^{c_\tau} - F_{\tau-1}^{c_{\tau-1}}$$

This difference can be decomposed in two parts:

$$p_\tau - p_{\tau-1} = (F_\tau^{c_\tau} - F_{\tau-1}^{c_\tau}) + (F_{\tau-1}^{c_\tau} - F_{\tau-1}^{c_{\tau-1}})$$

where the first term is a price difference when prices are evaluated with the same contract, and the second term is a calendar spread, i.e. the roll gap. At time τ we define the roll ratio as $\rho_\tau = \frac{F_{\tau-1}^{c_\tau}}{F_{\tau-1}^{c_{\tau-1}}}$ for $\tau \in \mathcal{R}$. We then construct the cumulative log adjustment as $A_t = \sum_{\tau > t, \tau \in \mathcal{R}} \log(\rho_\tau)$, which represents the cumulative roll ratios after time t, and notice that $A_T = 0$. The continuous log-price series is given by $q_t = \log(p_t) + A_t$. Let $r_t \equiv q_t - q_{t-1}$. Then, the back adjusted-price series in logs is:

$$\tilde{q}_t = \tilde{q}_{t-1} + r_t$$

with initial condition $\tilde{q}_0 = \log(p_0)$. This resulting back-adjusted price series does not display calendar change jumps.

As natural gas prices display seasonality, we then perform the STL seasonal adjustment of the price series $\tilde{q}_t$ for $t \in \{0, ..., T\}$. The resulting seasonally and back adjusted prices series (in logs) $\{\tilde{q}_t^{SA}\}_{t \in 0,...,T}$ is used as our front-month variable for the analysis.

In practice, we construct the seasonally and back adjusted prices series using 1-month (TZT1) and 2-month (TZT2) ahead futures contracts from Bloomberg, as 2-month ahead futures become front-month prices at roll dates. We repeat the same procedure using TZT13 and TZT14 price series to obtain the seasonally and back adjust 13-month ahead futures.

A.1.3 Oil forward demand proxy

The construction of the proxy IV for forward demand in the oil SVAR follows Gazzani, Venditti, and Veronese (2024). The UV selects the days on which there is a large change in stock market volatility, oil price volatility —measured by VIX and OVX, respectively— and the price of gold (New York COMEX price), and simultaneously a large change in front-month oil prices of the opposite direction. By large changes, we mean that, on the selected date, the change in the VIX, OVX, the price of gold, and the front-month oil price is below the sample's 10th percentile or above the 90th percentile. Using this method, we identify 25 events during our sample period listed in Table 4 below, with a brief description of the event. The Forward Demand IV takes the value of the VIX on the day when this large change occurs and 0 otherwise.

Table 4: Dates for which the proxy IV for the oil SVAR takes a value different from 0.

Date	**Event**	**VIX**	**OVX**	**Gold**
September 30, 2008	Rescue package hopes	−7.3	−4.6	−1.6
October 6, 2008	Global growth fears	6.9	8.6	4.0
October 24, 2008	Global growth fears	11.3	2.7	2.2
November 20, 2008	Global growth fears	6.6	7.5	1.7
January 20, 2009	Banking solvency fears	10.5	7.4	1.8
January 28, 2009	Bank bailout expectations	−2.6	−2.2	−1.3
February 17, 2009	Global growth fears	5.7	10.1	2.7
February 24, 2009	Banking health optimism	−7.1	−3.9	−2.6
March 3, 2009	Banking health optimism	−1.7	−3.5	−2.8
April 20, 2009	Global growth fears	5.2	4.7	2.2
May 6, 2010	Greek sovereign debt crisis	7.9	4.1	1.9

Continued on next page

Table 4 – *Continued from previous page*

Date	Event	VIX	OVX	Gold
August 8, 2011	U.S. sovereign downgrade	16.0	13.8	3.7
August 18, 2011	Euro sovereign crisis	11.1	14.8	1.5
September 7, 2011	Global growth rebound	−3.6	−2.5	−3.0
June 1, 2012	Global growth fears	2.6	6.8	3.6
January 5, 2015	Political uncertainty in Greece	2.1	3.4	1.5
January 12, 2015	Global growth fears	2.1	3.1	1.4
January 15, 2016	Global growth fears	3.1	4.5	1.6
February 11, 2016	Global growth fears	1.9	5.3	4.4
June 24, 2016	Brexit	8.5	3.2	4.6
April 2, 2018	Trade tensions	3.7	2.5	1.4
February 24, 2020	Coronavirus pandemic	8.0	3.7	1.7
May 5, 2023	Stronger-than-expected U.S. labor market	−2.9	−6.4	−1.5
April 10, 2025	Trade tensions	7.1	6.1	3.2
August 1, 2025	Trade tensions	3.7	2.6	1.6

Events are chosen based on the changes of VIX, OVX, gold prices, and front-month oil prices. For each date, a short reference to the events unfolding on that date are given. Until June 2020, many of the dates and titles come from Gazzani, Venditti, and Veronese (2024). For the dates absent in Gazzani, Venditti, and Veronese (2024), a Google News Search using Gemini is done to summarize the risk-on/risk-off events at the time. For each date, VIX, OVX, and gold returns are given. OVX, VIX indices are in simple differences. Gold returns are in log differences.

A.1.4 Total Degree Days

To proxy current natural gas demand, we construct a temperature-based instrument using daily mean temperature forecasts from the *CFSV2: NCEP Climate Forecast System Version 2* database (Saha et al., 2014). Temperatures are provided on a latitude-longitude grid and are aggregated to the country-level using population weights from the *LandScan Global Population Dataset* (Sims et al., 2023). Using these country-level temperatures, we further compute total

degree days (TDD) for each country. TDD measure weather-driven energy demand associated with both heating and cooling needs (National Weather Service, 2026). Let $T_{i,t}$ denote the daily mean temperature in country i. Heating degree days (HDD) and cooling degree days (CDD) are defined as

$$HDD_{i,t} = \max\{18 - T_{i,t}, 0\}$$

and

$$CDD_{i,t} = \max\{T_{i,t} - 18, 0\},$$

where temperatures are measured in degrees Celsius and 18°C (65°F) is the reference threshold commonly used in the literature. Total degree days are then computed as

$$TDD_{i,t} = HDD_{i,t} + CDD_{i,t}.$$

Higher values of $TDD_{i,t}$ indicate temperatures further away from the comfort threshold and therefore greater expected energy demand for either heating or cooling.

The country-level TDD series are aggregated into a single European TDD series using annual natural gas consumption weights from the *2025 Energy Institute Statistical Review of World Energy* (Energy Institute, 2026). We employ one-year lagged gas consumption weights. Specifically:

$$TDD_t = \sum_i w_{i,y(t)-1} TDD_{i,t},$$

where $y(t)$ denotes the calendar year corresponding to day t, and $w_{i,y(t)-1}$ denotes country i's share of total European natural gas consumption in the previous calendar year.

To further isolate unusual weather conditions from normal seasonal patterns, we construct temperature anomalies by comparing the European TDD series to its historical seasonal average. Let $\overline{TDD}_t^{(5)}$ denote the average value of TDD_t observed for the same calendar day over the previous five years. The temperature anomaly is defined as

$$A_t = TDD_t - \overline{TDD}_t^{(5)}.$$

Positive values of A_t indicate weather conditions associated with stronger-than-normal energy demand, while negative values indicate weaker-than-normal demand. The resulting

daily forecasted temperature anomaly series is used as an instrumental variable for current natural gas demand in the SVAR specification.

A.1.5 Sources

Table 5: Variables and data sources.

Variable name	**Source**	Transformation
Front-month oil price	Refinitiv (LCOc1)	See above
Dated Brent (spot Brent)	Argus Media	See above
TTF prices	Bloomberg (TZT1 Comdty, TZT2 Comdty, TZT13 Comdty, TZT14 Comdty)	See above
Global equity index	Refinitiv (ACWI)	One-year rolling window on daily returns
European equity index	Bloomberg (MXEU Index)	Same transformation as ACWI
GPR Threats index	https://www.matteoiacoviello.com/gpr.htm	Only working days selected
Proxy IVs		See above
OVX	Bloomberg (OVX Index)	None
VIX	Bloomberg (VIX Index)	None
Gold price (NY Comex price)	Bloomberg (GC1)	None
Consumer Price Index	Haver Analytics	Log
Unemployment rate	Haver Analytics	None
Global oil production	IEA	None
OECD oil inventories	IEA	None
Global oil inventories	Kpler	None
European gas inventories	GIE (Gas Infrastructure Europe)	Deviation from the 3-year average

This table lists the variables used in the high-frequency SVARs and the local projections, with their corresponding data sources and transformations.

A.2 SVAR Estimation

When using proxy IV, sign and relative magnitude restrictions for identification, the prior of the reduced-form VAR coefficients (B) in Equation 2 is a Normal-Wishart distribution characterized as:

$$B \mid \Sigma \sim \mathcal{N}(b_0, \Sigma \otimes \Phi_0) \tag{9}$$

$$\Sigma \sim \mathcal{IW}(S_0, \alpha_0) \tag{10}$$

and Φ_0 is a diagonal matrix where for any lag ℓ of variable j, its diagonal entry equals $\frac{1}{\sigma_j^2}\left(\frac{\lambda_1}{l^{\lambda_3}}\right)$. The prior's distribution parametrization used is the following: $b_0 = 0.8$ for each variable's own first lag and 0 otherwise, $\lambda_1 = 1$ and $\lambda_3 = 1$. σ_j^2 is pre-computed as the OLS residual variances of univariate $AR(p)$ regressions run on each endogenous variable. S_0 is a diagonal matrix with diagonal elements that come from the individual AR regressions, $\alpha_0 = n + 2$, where n is the number of variables in the SVAR, which equals 4.

For the measurement equation, Equation 3, BEAR follows Caldara and Herbst (2019) and imposes a prior $N(0, 1)$ for β and an inverse gamma distribution, $IG(s_1, s_2)$ for the standard deviation of the measurement error σ_v^2 with $s_1 = 2$ and $s_2 = 0.02$ for oil. For natural gas, we use the high relevance prior option in BEAR which imposes that half of the variation in the proxy can be attributed to measurement error.

The BEAR toolbox applies the Metropolis-within-Gibbs algorithm of Caldara and Herbst (2019). In this algorithm, the proxy IV pins down column b_1, the elements of the reduced-form matrix that correspond to the forward demand (or current demand for natural gas) shock, and the sign and magnitude restrictions enter in a second step inside the algorithm as rotation matrices. BEAR then checks the responses given by each iteration. If the sign and relative magnitude restrictions are respected, it keeps the draw, otherwise if rejects it. We simulate 2000 iterations, with the first 200 used as burn-in.

B Additional results

B.1 Impact on energy markets: Inventories

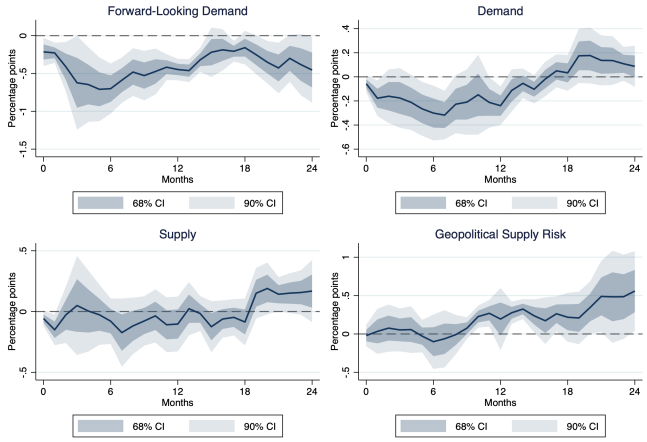


Figure 7: European gas storage responses to gas shocks

Notes: Responses to shocks normalized to raise the front-month oil price by 1 percent on impact. The shaded area corresponds to the 68-90% confidence interval.

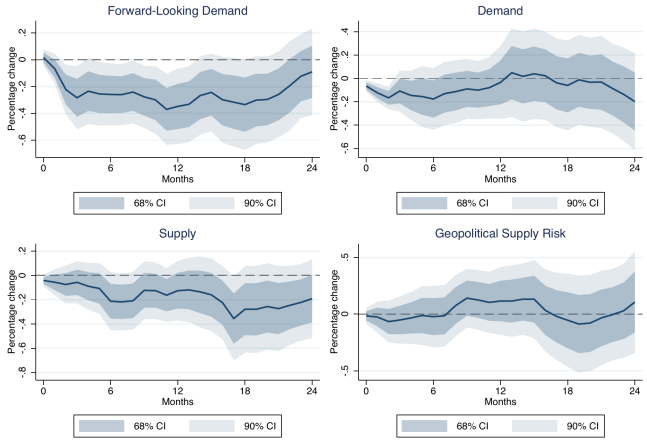


Figure 8: Global oil inventories responses to oil shocks

Notes: Responses to shocks normalized to raise the front-month oil price by 1 percent on impact. The shaded area corresponds to the 68-90% confidence interval.

B.2 Macroeconomic effects: Core consumer prices

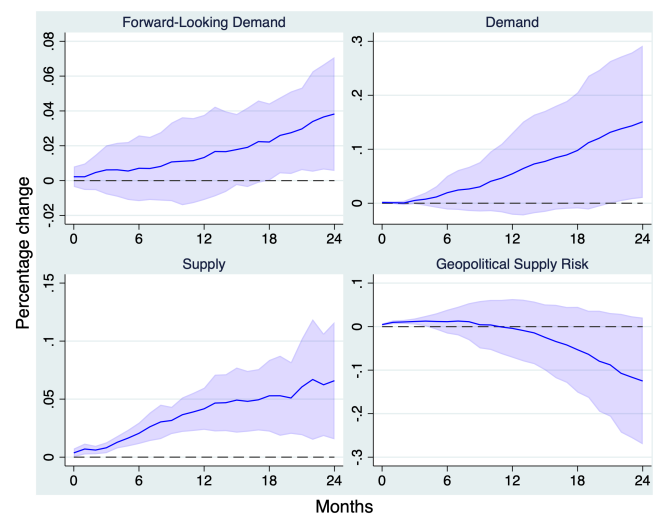
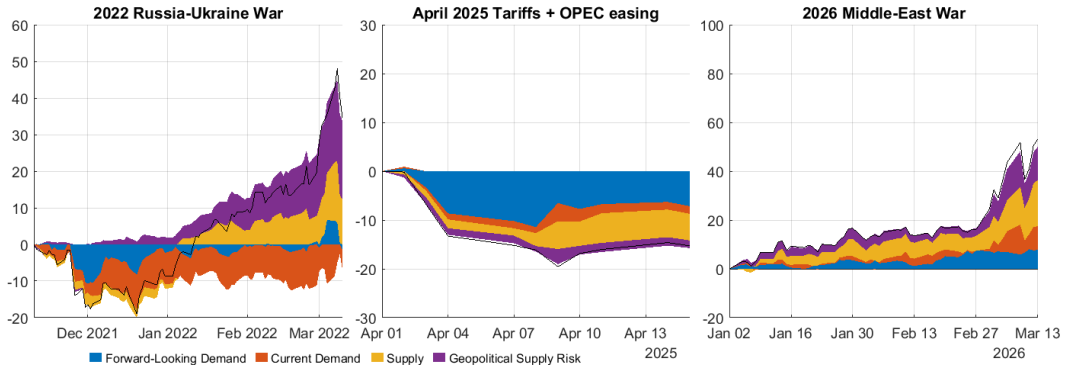


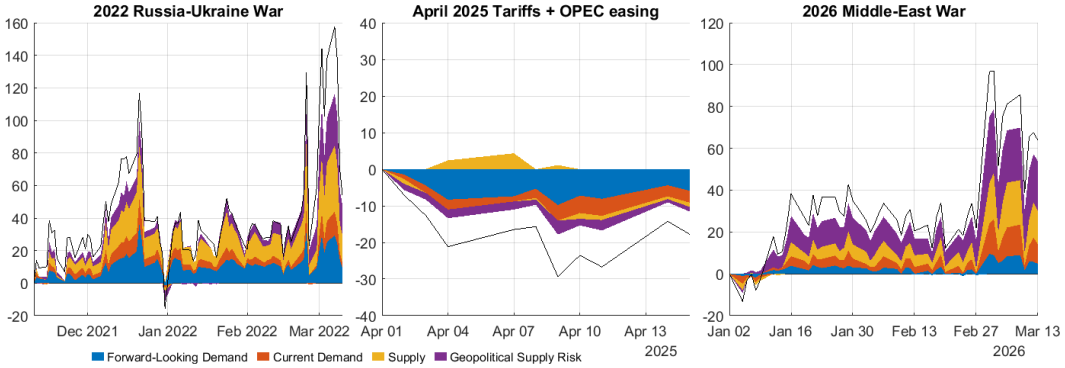
Figure 9: Impulse responses of core consumer prices to oil shocks.

Notes: Responses of consumer prices to shocks normalized to raise the front-month oil price by 1 percent on impact. Cumulative CPI log changes from t-1, multiplied by 100. Solid lines show estimated responses outside the COVID-19 window; shaded areas denote 90 percent pointwise confidence intervals. Horizons are in months.

B.3 Real-time historical decompositions



(a) Oil one-month price real-time decompositions



(b) Natural gas front-month real-time decompositions

Figure 10: Variance decompositions of front-month energy prices during three episodes

Notes: Variance decompositions of front-month prices energy prices using the four-variable SVAR models. Units in the horizontal axis are percent deviation from the starting point at the beginning of the period considered. Daily data data used to estimate the models are as follow. Oil starts on June 1, 2027; natural gas starts on October 4, 2013. The end dates in the real-time exercises are: (i) 2022 Russia-Ukraine war: March 10, 2023; (ii) April 2025 tariffs: April 15, 2025; (iii) 2026 Middle East War: March 13, 2026



PUBLICATIONS

Fear and Shortages: Identifying Geopolitical Risk and Supply Disruptions in Energy Markets
Working Paper No. WP/2026/218